

ROBERT ZHANG

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EDUCATION

University of California, Berkeley

Expected: May 2026

Bachelor of Science in Electrical Engineering and Computer Sciences

GPA: 4.0/4.0

- **Courses:** Data Structures, Algorithms, Probability, Operating Systems, Digital Design, Integrated Circuits, Machine Structures, Signals, Microelectronic Devices, Circuits, Linear Algebra

University of Minnesota, Twin Cities

Aug. 2022 – May 2023

Dual Enrollment in Engineering and Chemistry

GPA: 4.0/4.0

WORK EXPERIENCE

Software Development Engineer Intern, AWS

May 2025 – Aug. 2025

- Increased usable widget space by 20% by deploying accessibility features and creating a modernized user interface
- Engineered one-click deployment system for AWS account and widget resources using CDK and CloudFormation
- Developed website that uses Bedrock to automatically customize user widgets for 8x faster customer onboarding

Teaching Assistant, UC Berkeley Electrical Engineering and Computer Sciences

August 2024 – Present

- Teach and support 400+ students through weekly lab sections and office hours for linear algebra and circuits courses
- Develop new content for 5 labs and create assignments to transition from Jupyter Notebook to Marimo

Campus Ambassador, UC Berkeley Visitor Services

Sept 2023 – Present

- Provide engaging campus tours and manage the visitor center for thousands of students, families, and visitors

RESEARCH EXPERIENCE

Research Assistant, UC Berkeley Helen Wills Neuroscience Institute — Knight Lab

Aug. 2025 – Present

- Create a time-locking device for reliable electrophysiology readings from Responsive Neurostimulation tools
- Achieve reliable delays of with $<0.5\text{ms}$ jitter using an adjustable system of photodiodes and microcontrollers
- Train intracranial hyperscanning classification models to determine trends in human cooperation

Computational Researcher, UCSF Memory and Aging Center — Grinberg Lab

Jan. 2024 – June 2025

- Refactored segmentation pipeline for analyzing Tau proteins in Alzheimer's patients for 20% bandwidth increase
- Achieved 15x processing speed improvement by refining U-net segmentation architecture for larger image formats
- Improved segmentation accuracy by 12% through training a new TensorFlow model on GPU accelerated systems

Bioinformatics Analyst, University of Minnesota — Boshnakovski Lab

May 2022 – Aug. 2023

- Generated RNA sequencing data on over 2,000 Ewing sarcoma samples through the CHURP pipeline at MSI
- Conducted differential gene expression and KEGG pathway analysis to generate figures and tables for papers

ACTIVITIES

Course Coordinator and Mentor, Computer Science Mentors

Jan. 2024 – Present

- Lead team of 11 mentors, teach weekly sections, and develop targeted worksheets to reinforce course content

Activities Officer, Eta Kappa Nu (EECS Honor Society)

Jan. 2024 – Present

- Lead team of 7 officers to organize over 25 community events each semester and host tutoring and review sessions

Patient Amenity Volunteer, UCSF Volunteer Services

Jan. 2025 – Present

- Provide support and companionship to patients at UCSF Medical to improve their in-hospital experience

PROJECTS

PintOS and Distributed Systems | C, Rust, Assembly

Aug. 2025 – Present

- Optimize user processes, threading, and fast file system operations for the UNIX-based PintOS kernel
- Built a fault-tolerant MapReduce server to evenly distribute tasks across numerous workers

5-Stage RISC-V Processor | Verilog, FPGA, Vivado, VCS

Aug. 2025 – Present

- Code FPGA to execute instructions in a 5 stage pipeline with features like floating point unit, BIOS, and MMIO UART

Voice Controlled Vehicle | C++, Python, Breadboarding

Jan. 2024 – Apr. 2024

- Designed and constructed a voice-controlled vehicle with a closed-loop feedback system with 94% accuracy

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Python, C, C++, C#, R, Verilog, Assembly, SQL, CSS

Technologies: AWS, Docker, Linux, Git, LTSpice, GDB, Vivado

Libraries/Frameworks: React, Node.js, Next.js, pandas, NumPy, scikit-learn, Tensorflow, PyTorch, Matplotlib, dplyr, ggplot2